FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2015**

HALF COURSE TEST –VII

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-A (07 – 10)** contains 4 multiple choice questions having one or more than one correct answer and each question carries **+4 marks** for correct answer and there is no negative marking.

(iii) **Section-A (11 – 14)** contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** each for correct answer and **– 1 mark** for wrong answer.

(iv) **Section-A (15 – 23)** contains 3 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

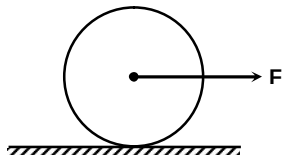
SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

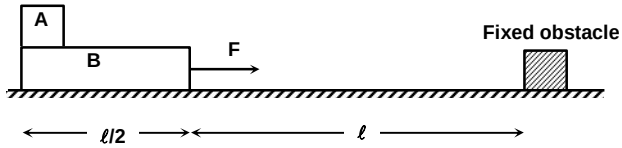
1. Velocity of a body of mass 2kg moving in x-y plane is given by $\vec{v} = \left(2\hat{i} + 4t\hat{j}\right)$ m/s, where t is the time in seconds. The power delivered to the body by the resultant force acting on it at t = 5 sec is
   (A) 80 Watts (B) 160 Watts
   (C) 40 Watts (D) 100 Watts

2. A solid sphere of mass m is lying at rest on a rough horizontal surface. The coefficient of friction between ground and sphere is μ. The maximum value of F, so that sphere will not slip, is equal to



   (A) $7\mu mg/5$ (B) $4\mu mg/7$
   (C) $5\mu mg/7$ (D) $7\mu mg/2$

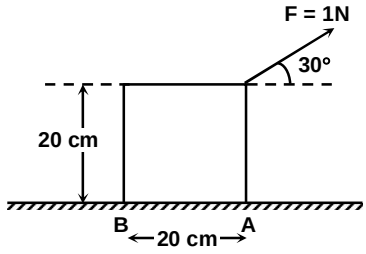
3. In the figure shown, mass of A and B is equal to M each. Friction between B and lowermost surface is negligible. Initially, both the blocks are at rest. The dimensions of the block A are very small.



A constant horizontal force F is applied on the block B and both the blocks start moving together without any relative motion.
Suddenly, the block B encounters a fixed obstacle and comes to rest. The block A continues to slide on the block B. The block A just manages to reach the opposite end of the block B. What is the coefficient of friction between the two blocks? (Required length are shown in the figure)
   (A) F/Mg (B) 2F/Mg
   (C) F/2Mg (D) none

Space for rough work

4. A block of mass $\sqrt{3}/10$ kg is placed on a rough horizontal surface as shown in the figure. A force of 1 N is applied at one end of the block and the block remains stationary. The normal force exerted by the surface on the block acts (g = 10 m/s^2)



(A) through the centre of mass of the block.
(B) through point *A*.
(C) through point *B.*
(D) through the point at a distance 5 cm. from *A*.

5. An insect of mass m is initially at rest at the one end of a stick of length L and mass M, which rests on a smooth horizontal floor. The coefficient of friction between the insect and stick is k. The insect starts moving to reach the other end in shortest possible time. Choose the incorrect statement.

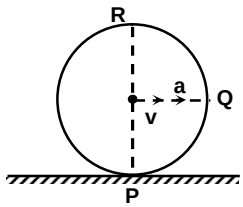
(A) The centre of mass of the stick has speed v = $\left[\frac{kmg}{M}\right]t$ w.r.t. horizontal floor at time t.

(B) Time taken by the insect to reach the other end is $t = \sqrt{\frac{2LM}{k(M+m)g}}$

(C) The magnitude of the linear momentum of the insect at time 't' is equal to kmgt

(D) The maximum speed acquired by the stick is $v_{max} = \sqrt{\frac{2kLMm^2g}{(M+m)m}}$

6. A uniform wheel is rolling without slipping on a horizontal surface. At a certain instant, its centre of mass has velocity 'v' and acceleration 'a'. P, Q and R are three points on the rim of the disc as shown in the figure. Choose the incorrect statement.



(A) Acceleration of P is vertically upwards
(B) Acceleration of Q may be vertically downwards
(C) Acceleration of R can be horizontal
(D) Acceleration of some point on the rim may be horizontal leftwards.

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. A particle moves with an initial velocity v_0 and retardation αv, where v is its velocity at any time t.
(A) The particle will cover a total distance v_0/α.
(B) The particle will come to rest after a time $1/\alpha$.
(C) The particle will continue to move for a very long time.
(D) The velocity of the particle will become $v_0/2$ after a time $1/\alpha$.

8. A wave is represented by the equation $y = A \sin (10\pi x + 15\pi t + \pi/3)$ where x is in meters and t is in seconds. The expression represents
(A) a wave travelling in the positive x-direction with a velocity 1.5m/s.
(B) a wave travelling in the negative x-direction with a velocity 1.5m/s.
(C) a wave travelling in the negative x-direction having a wavelength 0.2m.
(D) a wave travelling in the positive x-direction having a wavelength 0.2m.

9. A perfectly elastic uniform string is suspended vertically with its upper end fixed to the ceiling and the lower end is loaded with a weight. If a transverse wave is imparted to the lower end of the string, the pulse will
(A) not travel along the length of the string
(B) travel upwards with increasing speed
(C) travel upwards with decreasing speed
(D) travel upwards with constant acceleration

10. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficients of linear expansion of the two metals are α_c and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is
(A) proportional to ΔT
(B) inversely proportional to ΔT
(C) proportional to $|\alpha_B - \alpha_C|$
(D) inversely proportional to $|\alpha_B - \alpha_C|$

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT-1
Each molecule of a gas moves with rms speed if the temperature of gas is constant.
and

STATEMENT-2

The rms speed of molecules of a gas is equal to $\sqrt{\frac{3RT}{M}}$, where T and M are the temperature and molecular mass of the gas. R is the ideal gas constant.

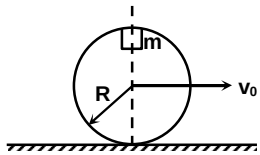
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

12. STATEMENT-1
Water is filled up in a hollow cylindrical container of conducting bases and adiabatic curved surface and kept vertical in an isolated system. If temperature of cylinder is decreased slowly from the bottom, the ice formation will start from the bottom.
and
STATEMENT-2
The temperature of liquid which is at the top will be lowest first.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

13. From the base of an incline plane inclined at an angle of 30° with the horizontal a particle is projected with a speed u. The angle of projection of the particle is 60° with the horizontal.
STATEMENT-1: The particle will strike the inclined plane normally for a particular u.
and
STATEMENT-2: The angle at which the particle hits the incline plane is independent of u.
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

14. STATEMENT-1: A ring of mass m and radius R has a mass m attached to it. The ring rolls without slipping on the horizontal surface as shown in the figure. At the instant shown the normal contact force exerted by the floor on the ring is less than (2mg).



and

STATEMENT-2: $\vec{F}_{ext} = m\vec{a}_{cm}$ (i.e. net external force acting on a rigid body = mass times acceleration of centre of mass)
(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

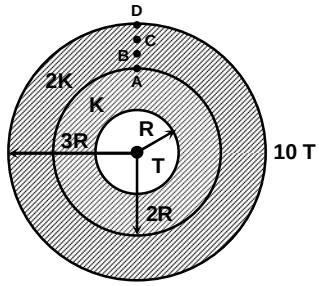
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 15 to 17

A composite spherical shell is made up of two materials having thermal conductivities K and 2K respectively as shown in the figure. The temperature at the innermost surface is maintained at T whereas the temperature at the outermost surface is maintained at 10T.



A, B, C and D are four points in the outer material such that AB = BC = CD. Now answer the following questions.

15. The effective thermal resistance between the inner surface of the shell and the outer surface of the shell for the radial heat flow is

(A) $\frac{1}{8\pi KR}$ (B) $\frac{1}{7\pi KR}$

(C) $\frac{7}{48\pi KR}$ (D) $\frac{6}{49\pi KR}$

16. The net rate of heat flow from the outermost surface to the innermost surface of the shell will be

(A) $63\pi RKT$ (B) $72\pi RKT$

(C) $\frac{432}{7}\pi RKT$ (D) $\frac{147}{2}\pi RKT$

17. Out of the segments AB, BC and CD the magnitude of the temperature difference will be maximum across

(A) AB (B) BC

(C) CD (D) equal across all the three.

Space for rough work

Paragraph for Question Nos. 18 to 20

Two coherent point light sources are placed at x = -d/2 and x = d/2. These sources are surrounded by a spherical screen having equation $x^2 + y^2 + z^2 = R^2$ (where R >> d). The sources emit monochromatic light of wavelength λ (= d/3).
Waves from both the sources reach screen and superimpose. When waves reaching any point on the screen have a constant path difference of $n\lambda$ (where λ is wavelength of light and n is some integer), the superposition of these waves is constructive in nature (i.e. maximum intensity) n may be called as order of maximum.

18. The number of maximas on the screen
(A) 7 (B) 12
(C) 6 (D) none of these

19. As screen is a bounded one, therefore the length of maxima is finite. Which of the following data is correct about shape and length of second order maxima?
(A) hyperbola, $\frac{2\pi R\sqrt{5}}{3}$ (B) circle, $\frac{2\pi R\sqrt{5}}{3}$
(C) hyperbola, $\frac{2\pi R}{3}$ (D) circle, $\frac{2\pi R}{3}$

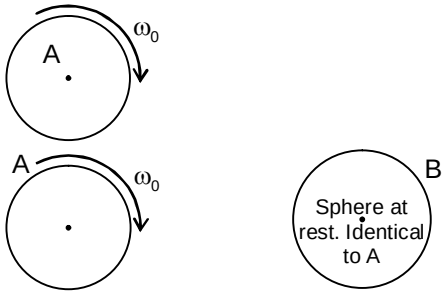
20. Which of the statements is incorrect about the interference experiment explained above
(A) if radius of spherical screen is doubled the intensity of maxima will become one fourth.
(B) All maximas are equally spaced along x-axis.
(C) The equation of first order maximum is $y^2 + z^2 = (8/9)\ R^2$ and x = R/3.
(D) none of these

Space for rough work

Paragraph for Questions Nos. 21 to 23

A uniform sphere of mass m and moment of inertia $\frac{2}{5}mR^2$ about its diameter is rotating with angular velocity ω_0 about a horizontal diametrical axis.

It is now placed on rough horizontal ground with friction coefficient μ.



21. The time taken by sphere A to achieve pure rolling is (assuming no collision)

(A) $\frac{R\omega_0}{\mu g}$ (B) $\frac{R\omega_0}{7\mu g}$

(C) $\frac{2R\omega_0}{7\mu g}$ (D) $\frac{R\omega_0}{4\mu g}$

22. Eventually, when sphere A collides elastically with sphere B, then immediately after collision the kinetic energy of sphere B is: (Assume no friction b/w spheres)

(A) $\frac{2}{49}\frac{mR^2\omega_0^2}{}$ (B) Zero

(C) $\frac{1}{49}mR^2\omega_0^2$ (D) $\frac{4}{49}mR^2\omega_0^2$

23. After collision, sphere B is allowed to move while sphere A is held back. The final kinetic energy of sphere B when pure rolling starts:

(A) $\frac{1}{2}mR^2\omega_0^2$ (B) $\frac{5}{343}mR^2\omega_0^2$

(C) $\frac{3}{343}mR^2\omega_0^2$ (D) $\frac{10}{343}mR^2\omega_0^2$

Space for rough work

Chemistry

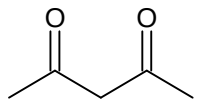
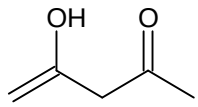
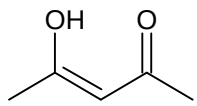
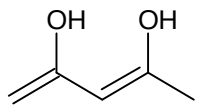
PART – II

SECTION – A

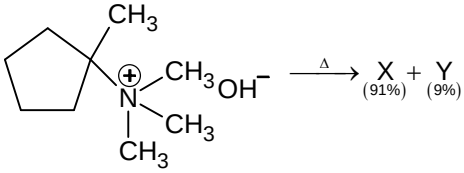
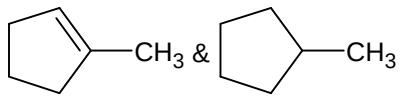
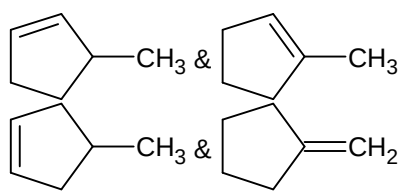
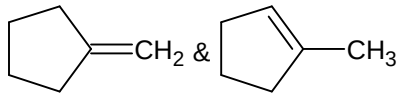
Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The catalyst decrease the E_a from 100 K mol^{-1} to 80 kJ mol^{-1}. At what temperature the rate of reaction in the absence of catalyst at 500 K will be equal to rate reaction in presence of catalyst.
(A) 400 K (B) 200 K
(C) 625 K (D) None of the above

2. Which structure is most stable?
(A) 
(B) 
(C) 
(D) 

3. Total number of stereoisomer of the compound 1 – bromo – 3 – chloro cyclobutane
(A) 0 (B) 1
(C) 2 (D) 3

4. Two alkenes, X (91% yield) & Y (9% yield) are formed when the following compound is heated

X and Y are
(A) 
(B) 
(C) 
(D)

Space for rough work

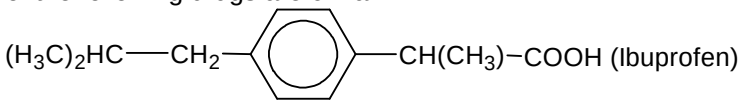
5. Potential energy of the electron at infinity is equal to
(A) 0 (B) ∞
(C) $-\frac{Ze^2}{R}$ (D) $-\frac{Z^2e}{R}$

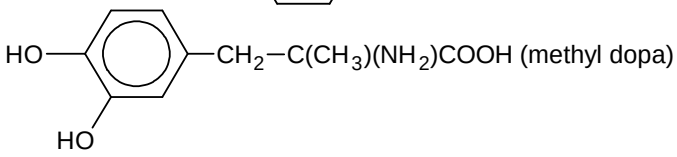
6. The rate law has the form
$\text{rate} = K[A][B]^{3/2}$
can the reaction be elementary process
(A) yes (B) no
(C) may be yes or no (D) cannot b predicted

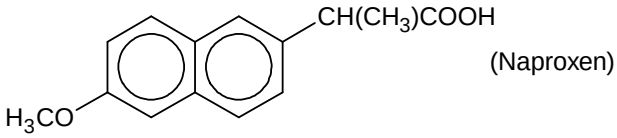
Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. Identify the correct statement (s) regarding hydrogen gas:
(A) unlike to H_2, D_2 do not show spin isomerism
(B) ortho hydrogen differs from para hydrogen w.r.t. their boiling point, specific heats & thermal conductivity
(C) as the temperature is increased from absolute zero some of the para form gradually changes to ortho form
(D) para hydrogen has lower energy than ortho hydrogen

8. Which of the following drugs are chiral?
(A) 

(B) 

(C) $(CH_3)_2C(SH)-CH(NH_2)COOH(\text{pencillamine})$
(D) 

9. Which of the following are intensive properties?
(A) temperature (B) density
(C) specific heat (D) molar volume

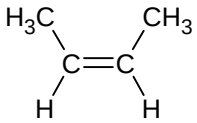
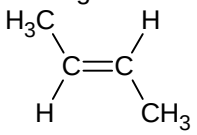
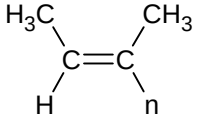
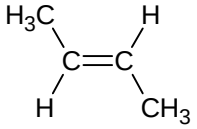
Space for rough work

10. Which of the following mixture represent a buffer?
(A) 100 ml of 2 M HCN & 100 ml of 1M NaOH
(B) 100 ml of 2 M CH_3COONa & 100 ml of 1 M HCl
(C) 100 ml of 0.2 M NH_4OH & 100 ml of 0.1 M H_2SO_4
(D) 100 ml of 0.5 M $C_6H_5NH_2$ & 100 ml of 0.2 M H_2SO_4

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT -1: CHF_3 is less acidic than $CHCl_3$
because
STATEMENT -2: Electron withdrawing groups increase the acidic nature.
(A) Statement -1 is True, Statement -2 is true; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is true; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

12. STATEMENT -1:

has higher boiling point than

but for melting point, it is the reverse.
because
STATEMENT -2:

has greater heat of hydrogenation than

(A) Statement -1 is True, Statement -2 is true; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is true; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Space for rough work

13. STATEMENT -1: A mixture of ideal gases is cooled up to liquid He temperature (4.22K) to form an ideal solution.
because
STATEMENT -2: An ideal gas cannot be liquefied as there exists no force of attraction among the gas molecules.
(A) Statement -1 is True, Statement -2 is true; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is true; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

14. STATEMENT -1: At equilibrium at unit activity, $\Delta G^0 = -RT \ln K_c$.
because
STATEMENT -2: Since $\Delta G = \Delta G° + RT\ln Q$ and at equilibrium, $\Delta G = 0$ & Q becomes K_c at unit activity.
(A) Statement -1 is True, Statement -2 is true; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is true; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Comprehension Type

This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 15 to 17

An organic Lewis acid (A) which gives fumes in moist air and intensity of fumes is increased when a rod dipped in NH_4OH is brought near to it. An acidic solution of (A) on addition of NH_4Cl and NH_4OH gives a precipitate (B) which dissolves in NaOH solution. An acidic solution of A does not gives precipitate with H_2S.

Read the above paragraph carefully and give the answer of following questions:

15. What will be structure of (A)?
(A) Planar
(B) Trigonal planar
(C) Pyramidal
(D) Octahedral

16. What will be (A) in above sequence?
(A) BF_3
(B) $AlCl_3$
(C) $GaCl_2$
(D) $ZnCl_2$

17. The intensity of fumes increase due to the formation of–
(A) NH_4OH
(B) HCl
(C) NH_4Cl
(D) $NaAlO_2$

Space for rough work

Paragraph for Question Nos. 18 to 20

Lithium only forms monoxide when heated in oxygen. Sodium forms monoxide & peroxide in excess of oxygen. Other alkali metals form superoxide with oxygen i.e. MO_2. The abnormal behaviour of Li is due to small size. The larger size of nearer alkali metals also decides the role in formation of superoxides. The three ions are related to each other as follows:

$$\underset{\text{(Oxide ion)}}{O^{2-}} \xrightarrow{\frac{1}{2}O_2} \underset{\text{(Peroxide ion)}}{O_2^{2-}} \xrightarrow{O_2} \underset{\text{(Superoxide ion)}}{2O_2^-}$$

All the three ions abstract proton from water.

18. Consider the following reaction:

$$M + O_2 \longrightarrow \underset{\text{(superoxide)}}{MO_2} \quad (M = \text{alkali metal})$$

Select the correct statement:

(A) M cannot be Li and Na
(B) M cannot be Cs & Rb
(C) M cannot be Li & Rb
(D) none of these

19. Lithium does not form stable peroxide because

(A) of its small size
(B) d – orbitals
(C) it is highly reactive & form superoxide in place of peroxide
(D) covalent nature of peroxide

20. Which anion is stable towards water?

(A) O^{2-}
(B) O_2^{2-}
(C) O_2^-
(D) none of these

Space for rough work

Paragraph for Question Nos. 21 to 23

A reversible reaction proceeds in the direction that decrease the free energy of the system. There is an important relationship between the equilibrium constant 'K' for a reaction and the standard free energy (ΔG^0): $\log K = \frac{-\Delta G^0}{2.30\ RT}$

This equation is application to all reversible processes, allows us to estimate K form standard free energy of the reaction.

The value of equilibrium constant of an endothermic reaction rises with temperature, whereas a temperature increase lowers the equilibrium constant of an exothermic reaction.

$$\log\left[\frac{k_2}{k_1}\right] = \frac{\Delta H^0}{19.1}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

If ΔH^0 is positive (endothermic reaction) and $T_2 > T_1$ then $\log\frac{k_2}{k_1}$ is positive. The opposite situation occurs if ΔH^0 is negative.

21. The equilibrium constant for the dissociation of N_2O_4 into NO_2 is 0.155 at 25°C. What is the standard free energy change for this reaction given (log 0.155 = –0.810)
(A) –4.61 kJ/mole (B) +4.61 kJ/mole
(C) +8.12 (D) –8.12

22. The standard Gibbs free energy change for the reaction:
$2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$
is +11.70 kJ/mole at 25°C. What is the equilibrium constant for the reaction at 25°C?
(A) 8.8×10^{-3} (B) 1.6×10^{-4}
(C) 2.6×10^{-3} (D) 4.2×10^{-5}

23. The heat of formation of ammonia is –46.2 KJ/mole. What is the equilibrium constant for the reaction:
$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$
at 500°C? The equilibrium constant at 25°C is 6.8×10^5 (given antilog (–9.98) is equal to 1.1×10^{-10})
(A) 2.5×10^{-4} (B) 5.0×10^{-5}
(C) 7.5×10^{-5} (D) 1.0×10^{-6}

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Let $f(x) = e^{\{e^{|x|}\operatorname{sgn} x\}}$ and $g(x) = e^{[e^{|x|}\operatorname{sgn} x]}, x \in R$ where {x} and [.] denotes the fractional part and integral part functions respectively. Also $h(x) = \ln(f(x)) + \ln(g(x))$ then for all real x, h(x) is
   (A) an odd function (B) an even function
   (C) neither an odd nor an even function (D) both odd as well as even function

2. If a, b, c are three complex numbers selected randomly (without repetition) from the set {1, 2, 3 ….. 2013) then the probability that $\arg\left(\frac{a-b}{a-c}\right) = 0$ is
   (A) $\frac{1}{3}$ (B) $\frac{2}{3}$
   (C) $\frac{3}{2013}$ (D) none of these

3. Let a, b, c $\in$ R be such that a + b + c < 0, a – b + c < 0 and c > 0, if 'α' and 'β' are roots of the equation $ax^2 + bx + c = 0$, then value of $[\alpha] + [\beta]$ is
   (A) 2 (B) 1
   (C) –1 (D) 0

4. If a – b, b – c, c – a are in A.P., then the straight line (a – b)x + (b – c)y + (c – a) = 0 will pass through
   (A) (1, 2) (B) (2, 1)
   (C) (2, 3) (D) (3, 1)

Space for rough work

5. If $x = \log_e(\cos\theta), y = \cos(\log_e\theta)$ where $e^{-\pi/2} < \theta < e^{\frac{\pi}{2}}$, then
(A) x = y
(B) x > y
(C) x < y
(D) none of these

6. Evaluate $3^{2-\log_3 4}$
(A) $\frac{4}{9}$
(B) $\frac{9}{4}$
(C) 36
(D) none of these

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. The function $f(x) = x^{\frac{1}{\ln x}}$
(A) is a constant function
(B) has a domain $(0,1)\cup(e,\infty)$
(C) is such that $\lim_{x\to 1} f(x)$ exist
(D) is periodic

8. If $\log_2(3^{2x-2}+7) = 2+\log_2(3^{x-1}+1)$ then x equals
(A) 0
(B) 1
(C) 2
(D) none of these

9. If $\frac{2}{1!13!}+\frac{2}{3!11!}+\frac{2}{5!9!}+\frac{1}{7!7!} = \frac{2^m}{n!}$ then
(A) m + n = 27
(B) $m^2 = 1 + 2n$
(C) $n^2 - m^2 = 27$
(D) n = 1 + m

10. If $\left(8+3\sqrt{7}\right)^n = P + F$, where P is an integer and F is a proper fraction, then
(A) P is an odd integer
(B) P is an even integer
(C) F.(P + F) = 1
(D) (1 – F)(P + F) = 1

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT 1: $P(A \cap B \cap C) = P\left(\frac{A}{B}\right).P\left(\frac{B}{C}\right).P(C)$

STATEMENT 2: $P\left(\frac{A}{B}\right)$ is probability of A when B has already happened

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

12. STATEMENT 1: If $x + y + z = xyz$, then at most one of the number can be negative
STATEMENT 2: In a triangle ABC, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ and there can be at most one obtuse angle in a triangle
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

13. STATEMENT 1: Diagonals of any parallelogram inscribed in an ellipse always intersect at the centre of the ellipse
STATEMENT 2: Centre of the ellipse is the only point at which two chords can bisect each other and every chord passing through the centre of the ellipse gets bisected at the centre
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

14. STATEMENT 1: The number of non-negative integer solutions to the inequality $x + y + z \leq 2013$ is ${}^{2016}C_3$

STATEMENT 2: $\sum\left({}^{n+1}C_r - {}^{n}C_r\right) = \sum\left({}^{n}C_{r-1}\right)$

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 15 to 17

Read the following write up carefully and answer the following questions:

An equation $a_0 + a_1x + a_2x^2 + ... + a_{99}x^{99} + x^{100} = 0$ has roots ${}^{99}C_0$, ${}^{99}C_1$, ${}^{99}C_2$, ... ${}^{99}C_{99}$

15. The value of a_{99} is equal to
(A) 2^{98} (B) 2^{99}
(C) -2^{99} (D) none of these

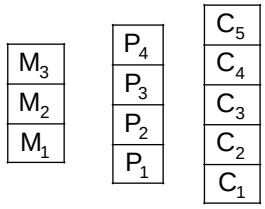
16. The value of a_{98} is
(A) $\frac{2^{198} - {}^{198}C_{99}}{2}$ (B) $\frac{2^{198} + {}^{198}C_{99}}{2}$
(C) $2^{99} - {}^{99}C_{49}$ (D) none of these

17. The value of a_{97} is
(A) $\frac{2^{198} - {}^{198}C_{99}}{2}$ (B) $\frac{2^{198} + {}^{198}C_{99}}{2}$
(C) $2^{99} - {}^{99}C_{49}$ (D) none of these

Space for rough work

Paragraph for Question Nos. 18 to 20

Read the following write up carefully and answer the following questions:
There are three different lots of Mathematics, Physics and Chemistry books arranged in three different lots as shown in the following diagram



18. In how many distinct ways we can pile up these books in a single lot without any restriction

(A) $\frac{12!}{3!}$ (B) $12!$

(C) $\frac{12!}{4!}$ (D) $\frac{12!}{5!}$

19. In how many distinct ways we can pile up these 12 books such that at time of piling up all these books in a single lot we always pickup the book from the top of respective lots

(A) $\frac{12!}{3!\times 5!}$ (B) $\frac{12!}{4!\times 5!}$

(C) $\frac{12!}{3!\times 4!\times 5!}$ (D) $\frac{12!}{3!\times 4!}$

20. In how many different ways we can pile up such that no two physics book are together

(A) ${}^9P_4 \times 8!$ (B) ${}^8P_4 \times 8!$

(C) ${}^8P_4 \times 9!$ (D) ${}^9P_4 \times 9!$

Space for rough work

Paragraph for Question Nos. 21 to 23

Read the following write up carefully and answer the following questions:

Given the equation of a hyperbola in general 2nd degree form i.e. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we can obtain the equation of its pair of asymptotes by adding a constant λ to the above equation. Now since after adding the constant λ, the new equation represents a pair of straight lines, therefore the determinant $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c+\lambda \end{vmatrix}$ vanishes to zero. Hence λ and the equation of pair of asymptotes can be found. If C represents the equation of a conjugate hyperbola, then C + H = 2A, where H represents the equation of hyperbola and A the equation of pair of asymptotes

21. The equation of a hyperbola conjugate to the hyperbola $x^2 + 3xy + 2y^2 + 2x + 3y = 0$ is
(A) $x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$ (B) $x^2 + 3xy + 2y^2 + 2x + 3y + 2 = 0$
(C) $x^2 + 3xy + 2y^2 + 2x + 3y + 3 = 0$ (D) $x^2 + 3xy + 2y^2 + 2x + 3y + 4 = 0$

22. A hyperbola passing through origin has $3x – 4y –1 = 0$ and $4x – 3y – 6 = 0$ as its asymptotes, then the equation of its transverse and conjugate axes are
(A) $x – y – 5 = 0$ and $x + y + 1 = 0$ (B) $x – y = 0$ and $x + y + 5 = 0$
(C) $x – y – 5 = 0$ and $x – y – 1 = 0$ (D) $x + y – 1 = 0$ and $x – y– 5 = 0$

23. The tangent at any point of a hyperbola $16x^2 – 25y^2 = 400$ cuts off a triangle from the asymptotes and that the portion of it intercepted between the asymptotes, then the area of this triangle is
(A) 10 sq. units (B) 20 sq. units
(C) 30 sq. units (D) 40 sq. units

Space for rough work